MA11003: Advanced Calculus

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1. Introduction

My typeset notes to our MA11003: Advanced Calculus course, taught in the Autumn Semester, 2025. For a more complete set of notes I will refer you to CSE2020. I am told *Advanced Engineering Mathematics* by Kreyszig is also good. The headings are the same as in the syllabus. Proofs of theorems were not part of the course, but I will give a few that I find are interesting enough.

2. Differential Calculus

2.1. Continuity Theorems

I don't want to go too deep into analysis; this is an introductory engineering course! I will note that these proofs depend on a critical property of the reals, called *completeness*. Proofs of these given without first exploring the least upper bound property and sequences are quite uninspiring.

IVP and EVT

Intermediate Value Theorem : If f is continuous on $[a, b]$, then for all $m \in (f(a), f(b))$, there exists $l \in (a, b)$ such that $f(l) = m$. Alternatively, f maps intervals to intervals.

Extreme Value Theorem: A continuous function on a closed set attains a maximum and minimum.

2.2. Mean Value Theorems

Rolle's, Cauchy's, Lagrange's and Taylor's Theorems

Rolle's Theorem: If f is continuous on $[a, b]$ and differentiable on (a, b), with $f(a) = f(b)$ then there exists $c \in (a, b)$ such that $f'(c) = 0$.

Lagrange's Mean Value Theorem (MVT): If f is continuous on $[a, b]$ and differentiable on (a, b), then there exists $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Cauchy's Mean Value Theorem (CMVT): If f, g are continuous on $[a, b]$ and differentiable on (a, b), then there exists $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{g(b) - g(a)}.$$

Taylor's Theorem: If f is differentiable $n + 1$ times on (a, x), with $f^{(n)}$ continuous on $[a, x]$ then there exists $c \in (a, b)$ such that

$$f(x) = \underbrace{\sum_{i=0}^{k} \frac{f^{(i)}(a)}{i!}(x - a)^i}_{\text{Taylor Polynomial}} + \underbrace{\frac{f^{(n+1)}(c)}{(n + 1)!}(x - a)^{n+1}}_{\text{Remainder Term}}$$

These are called *mean value theorems* since they relate to averages. Taylor's theorem is a polynomial version of MVT, hence included here. These theorems have geometrical interpretations. For instance MVT says that there is a tangent parallel to the secant connecting $f(a)$ and $f(b)$.

Rolle's Theorem

We initially have the intuition that the curve "flattens" whenever $f' = 0$. But this is false: consider $f(x) = |x|^3$ at $x = 0$. However notice that minima and maxima do have the property we want. So we propose the following:

Claim. Let c be a point such that $f(c) = \max_{a \leq x \leq b}(f(x))$. Then $f'(c) = 0$.

Proof. By the definition of the derivative,

$$f'(c) = \underbrace{\lim_{h \to 0^+} \frac{f(c + h) - f(c)}{h}}_{\leq 0} = \underbrace{\lim_{h \to 0^-} \frac{f(c + h) - f(c)}{h}}_{\geq 0}$$

The numerator is always negative by maximality of $f(c)$. Since $f'(c) \leq 0$ and $f'(c) \geq 0$, $f'(c) = 0$. A similar claim may be shown for the minima of f.

Now we would like to apply **EVT**. However the function could attain its extrema on the endpoints. To deal with this, pick an x such that $f(x) \neq 0$. If no such x exists then the theorem is trivially true.

f will have an extremum at c by **EVT** in (a, b) (since $f(x) \neq 0$, we are guaranteed to hit either a minima or a maxima). Now by our claim, the proof is finished.

Mean Value Theorem

A more comprehensive discussion of the MVT may be found here.

The idea is to try to use what we already know to prove the MVT.

$$f'(c) = \frac{f(b) - f(a)}{b - a} \Longleftrightarrow f'(c) - \frac{f(b) - f(a)}{b - a} = 0$$

So we will consider $g(x) = f(x) - x\frac{f(b)-f(a)}{b-a}$. I will leave it as an exercise to verify **Rolle's theorem** on g, and to similarly prove **CMVT**.

Taylor's Theorem (REDO)

Vaguely speaking, Taylor's theorem is a polynomial mean value theorem. Rearranging MVT we get $f(x + h) = f(x) + hf'(x + th)$ for some $t \in (0, 1)$.

Consider a polynomial p of degree $n + 1$ that satisfies:
- $p^{(i)}(x) = f^{(i)}(x)$ for $i = 0, 1, ..., n$
- $p(x + h) = f(x + h)$

Consider $g(x) = f(x) - p(x)$. Now $g^{(i)}(x) = 0$ for $i = 0, 1, ..., n$ and $g(x + h) = 0$. Now by repeated application of Rolle's, $g^{(n+1)}(x + th) = 0 \Rightarrow f^{(n+1)}(x + th) = \text{constant}$.

Example 1

Using **CMVT**, show that $\sin x < x$ on $(0, \frac{\pi}{2})$.

$$\frac{\sin x}{x} \overset{\text{CMVT}}{=} \cos c < 1$$

Example 2

Using **Taylor's theorem**, calculate $e^{0.1}$ to 1% accuracy.

Using $c < 0.1$,

$$1 < e^{0.1} = 1 + 0.1 + \frac{c^2}{2} \le 1.015.$$

If we can analyse the remainder term, we can analyse the error in our approximation.

2.3. Functions of Many Variables

We will not dive too deep into metric spaces and open sets and such.

Neighborhoods

We will define the distance between two points the usual way:

$$d((x_1, y_1), (x_2, y_2)) = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

$$d(x_1, x_2) = |x_1 - x_2|$$

The ε neighborhood of a point (a, b) is the set of all points that satisfy $d((x, y), (a, b)) \leq \varepsilon$.

The deleted neighborhood of a point does not include the point itself

Limits

We say

$$\lim_{(x_0, y_0) \to (x, y)} f(x, y) = L$$

if for all $\delta > 0$ there exists $\varepsilon > 0$ such that $|f(x, y) - L| < \varepsilon$ whenever $\sqrt{(x - x_0)^2 + (y - y_0)^2} < \delta$, that is, any point in the deleted δ neighborhood of (x_0, y_0) gets mapped onto the interval $(L - \varepsilon, L + \varepsilon)$.